

# Which Damage Do the Images Support? Evidence-Bounded 3D Damage Estimation on Turbine Blades

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## Abstract

001 Visual inspection of aero-engine blades must decide  
002 whether a change seen in an image is physical damage,  
003 where it is on the part, and how deep it goes. Ex-  
004 isting methods flag anomalies in 2D, output boxes or  
005 voxels, or require a measured 3D scan, and none states  
006 whether the available views are sufficient to support  
007 a claimed shape change. We study evidence-bounded  
008 3D damage estimation: given a healthy CAD model  
009 and a few calibrated images, output a displacement  
010 field on the blade surface when the images support one,  
011 and abstain when camera error, lighting or a surface-  
012 colour change explain the images equally well. Our  
013 estimator compares a nuisance-only explanation (per-  
014 view pose, light gains, an albedo patch) with the same  
015 explanation plus a continuous dent, using a differen-  
016 tiable renderer with a neural shader. Dent hypotheses  
017 are proposed by a Rao score test computed for every  
018 surface location in one backward pass, and the deci-  
019 sion threshold on the resulting fit gain is calibrated on  
020 healthy and colour-only blades with a split-conformal  
021 guarantee. We release BladeDelta, a benchmark of 696  
022 rendered inspection cases on 84 public Rotor37 blade  
023 geometries, with exact damaged meshes, colour-only  
024 negatives, three dent families, controlled capture per-  
025 turbations and two sealed test splits, plus 456 cases on  
026 six industrial blade designs. On a sealed test of 30 un-  
027 seen blades the method locates 54% of visible dents (12  
028 of 13 deeper than 0.2 mm; median depth error 0.03 mm)  
029 at a 10% false-alarm rate, while CAD-prior deforma-  
030 tion fields and a lifted 2D detector locate at most 16%;  
031 a second sealed test replicates the result (54% found,  
032 6% false alarms). On the unseen industrial designs,  
033 correcting the healthy reference for silhouette and cam-  
034 era motion raises AUROC from 0.72 to 0.95 and cuts  
035 false alarms from 51% to 4%.

## 1. Introduction

Turbine and compressor blades are inspected with cam-  
eras, and the inspector’s question is physical: is there  
a dent, where on the blade is it, and how deep is it?  
Damage of interest is small. On a blade imaged at  
 $512 \times 512$  pixels, a dent of 0.07 mm changes a few hun-  
dred pixels by two to seven grey levels. A one-degree  
camera error, a ten-percent lighting change or a stain  
changes thousands.

Current vision methods answer an easier question.  
Industrial anomaly detection localises unusual pix-  
els [1, 18], including under pose and illumination  
change [3, 11]. Reference-based methods compare a  
photograph with a CAD model and output boxes [2], or  
lift per-view scores into an anomaly volume [7]. CAD-  
conditioned reconstruction fits a deformation field to  
sparse posed views [17, 21], and leaves measurement-  
quality damaged surfaces to future work [17]. None  
of these outputs states whether the images actually  
support a physical change, as opposed to a nuisance  
explanation.

Our own earlier experiments made this concrete. A  
render-and-compare fit on public Rotor37 blade geom-  
etry produced false dents on 14 of 14 healthy or colour-  
only blades under a one-degree camera error, and on  
unfamiliar dents the best image match selected a worse  
3D shape in 8 of 12 depth sweeps. Better image agree-  
ment is not better geometry, and a plausible recon-  
struction can choose the wrong physical explanation.

We therefore formulate evidence-bounded 3D dam-  
age estimation. The input is a healthy CAD mesh and  
 $K$  calibrated images; the output is a per-vertex dis-  
placement field together with a decision, supported or  
not supported by these views. Our estimator treats  
this as a nested hypothesis comparison.  $H_0$  explains  
the images with nuisance only: per-view pose correc-  
tions, light gains and a surface albedo patch.  $H_1$  adds a  
continuous dent. The fit gain  $L(H_0) - L(H_1)$  is thresh-  
olded at a value calibrated on negatives, giving a con-

075 trolled false-change rate.

076 Making this work required separating the roles of  
077 the evidence. Pose is fitted from silhouettes, which do  
078 not depend on lighting; the dent is compared against  
079 nuisance only on interior surface pixels that are not  
080 grazing, not near a silhouette and not on the blade  
081 root. Dent hypotheses come from a Rao score test  
082 evaluated for every surface location and radius with  
083 one backward pass through the renderer, and the dent  
084 and colour-patch explanations are searched symmetrically,  
085 so a colour patch fitted first cannot hide a dent.  
086 We also report observability: on our development split,  
087 70% of dents placed on camera-facing surfaces are less  
088 than half visible in the two input views, and the correct  
089 output for those is to abstain.

090 Contributions:

- 091 • The task of evidence-bounded 3D damage estimation  
092 from sparse views and healthy CAD, with metrics  
093 that separate localisation, shape accuracy, false  
094 physical change and abstention on unobservable  
095 damage.
- 096 • An estimator combining a differentiable renderer  
097 with a neural shader, silhouette-only pose, interior-  
098 domain nested hypothesis testing, score-test proposals  
099 and split-conformal calibration.
- 100 • BladeDelta, a benchmark built on public Rotor37  
101 geometry [4] with exact damaged meshes, colour-  
102 only negatives, three dent families that differ from  
103 the estimator’s own profile, and controlled camera  
104 and lighting perturbations, with a protocol and  
105 sealed test split fixed before evaluation.

## 106 2. Related Work

107 Industrial and 3D anomaly detection. Memory-bank  
108 methods such as PatchCore [18] localise anomalous pixels;  
109 MVTEC 3D-AD [1] and Anomaly-ShapeNet [13]  
110 add measured or synthetic 3D. Pose-agnostic settings  
111 use multi-pose benchmarks [23] and Gaussian-splat  
112 references [11], and M<sup>2</sup>AD [3] benchmarks view-  
113 illumination interplay. Multi-view settings include  
114 RAD [5] and SiM3D [7], which lifts per-view scores into  
115 an anomaly volume; recent work warns that adding  
116 views can leak information in multi-view evaluation [6].  
117 Synthetic anomalies have been argued sufficient for 3D  
118 detection [20], and incomplete-modality methods train  
119 with 3D and test on RGB [19]. These methods output  
120 anomaly scores; they do not estimate a physical surface  
121 change or state whether the views support one.

122 Reference CAD and blade inspection. Looking 3D [2]  
123 aligns a query photograph with a reference shape and  
124 localises abnormal regions with boxes. BladeSynth [8]  
125 renders procedural nicks, dents, scratches and corrosion

on aero-engine blades for 2D detection. Model-free universal  
detection reconstructs a normal shape by GAN inversion [15].  
We instead output a 3D displacement field on a known design.

CAD-conditioned reconstruction. CAD-NeRF [21] retrieves  
a CAD model and jointly optimises deformation, pose and  
appearance from few uncalibrated views. CADSplat [17]  
fits a shared deformation field to sparse posed views with  
a CAD prior and explicitly leaves measurement-quality  
deformed surfaces to future work. Our free-form deformation  
baseline follows this family. The difference is the output:  
a decision on whether the deformation is supported, with a  
calibrated false-change rate.

Inverse rendering and uncertainty. Inverse rendering  
separates shape and material from images [16, 22]; we  
use differentiable rasterisation [12] with a learned  
shader for the same purpose on a known geometry. Radiance-  
field uncertainty [9, 10] and observation completeness [14]  
quantify where a reconstruction is unreliable. They do not  
test a physical-change hypothesis against nuisance explanations,  
which is the question an inspector needs answered.

## 511 3. Method

Setup. We are given a healthy mesh  $\mathcal{M} = (V, F)$  with  
vertices  $V \in \mathbb{R}^{N \times 3}$ ,  $K$  observed images  $I_1, \dots, I_K$  with  
nominal intrinsics and extrinsics, and nominal lighting.  
We seek a displacement field  $D \in \mathbb{R}^{N \times 3}$  on the airfoil  
and a decision  $s \in \{\text{supported}, \text{not supported}\}$ . Nuisance  
variables  $\eta$  are a per-view pose correction (a rotation about  
the part and a camera-frame translation), three light gains,  
and an optional albedo patch.

Differentiable renderer. We rasterise  $\mathcal{M} + D$  with  
nvd-iffrastr [12] using smooth vertex normals. A neural  
shader maps the per-pixel world normal  $n$ , position  $x$  and  
view direction  $\omega$  to one radiance component per light,  
 $R_k(n, x, \omega)$ , and an environment term  $R_0$ . Because light  
transport is linear in each light’s intensity, the pixel colour  
is  $T(a \odot (R_0 + \sum_k g_k R_k))$  for albedo multiplier  $a$ ,  
light gains  $g$  and a learned monotone tone curve  $T$ , followed  
by anti-aliasing and the sensor’s pixel filter. An unstructured  
shader that took  $g$  as an input learned to ignore it, and  
lighting changes then produced spurious dents. The shader  
is trained once, on renders of blade geometries disjoint from  
all evaluation geometries, with a paired loss on image  
differences between lighting settings. The healthy CAD  
rendered under the nominal capture provides a per-pixel  
correction

174  $B = I^{\text{ref}} - \hat{I}(\mathcal{M}; \eta_0)$  that absorbs static shader error.  
 175 On a held-out blade its mean error is 0.92 grey levels;  
 176 image changes caused by a lighting change correlate  
 177 with the reference renderer at 0.90–0.98, and changes  
 178 caused by a dent at 0.96–0.98 in the 120° view.

179 Nuisance fit ( $H_0$ ). Each view’s pose is initialised by  
 180 Euclidean ECC alignment of the observed and refer-  
 181 ence images, then refined by matching anti-aliased sil-  
 182 houettes to the observed foreground, coarse to fine. Sil-  
 183 houettes do not depend on lighting, which prevents the  
 184 pose from compensating for a brightness change. With  
 185 pose fixed, light gains are fitted on the full image.

186 Evidence domain. All further comparisons use a fixed  
 187 interior domain  $\Omega$ : airfoil pixels at least 6px from any  
 188 silhouette or self-occlusion edge, excluding the blade  
 189 root and pixels where  $|n \cdot \omega| < 0.35$ . Both hypothe-  
 190 ses are evaluated on the same  $\Omega$ , so their losses are  
 191 comparable.

192 Dent model ( $H_1$ ). A dent is parameterised by a sub-  
 193 vertex centre  $c$  (an anchor vertex plus a tangent offset),  
 194 depth  $\delta \geq 0$ , radius  $r$ , aspect ratio, orientation, profile  
 195 exponent  $p$  and a signed rim coefficient  $\kappa$ :

$$196 \quad D_i = -\delta \left[ (1 - s_i^2)_+^p - \frac{\kappa}{2} \rho(s_i) \right] n_c, \quad (1)$$

197 where  $s_i$  is the anisotropic normalised distance of ver-  
 198 tex  $i$  from  $c$  in the tangent plane,  $\rho$  a ring profile and  $n_c$   
 199 the centre normal, restricted to same-side airfoil ver-  
 200 tices. None of the benchmark’s truth families matches  
 201 this profile exactly.

202 Score-test proposals. Let  $h \in \mathbb{R}^N$  be an inward nor-  
 203 mal displacement. One backward pass gives  $g =$   
 204  $-\partial L_\Omega / \partial h$  at  $h = 0$ . For a kernel  $w_{c,r}$  centred at  $c$   
 205 with radius  $r$ , the normalised score  $\langle w_{c,r}, g \rangle / \|w_{c,r}\|$  is  
 206 the Rao score statistic for a small dent at  $(c, r)$  against  
 207  $H_0$ . We evaluate it for all candidate centres and two  
 208 radii with a sparse matrix product, and start  $H_1$  fits  
 209 from the top- $k$  after non-maximum suppression. Pro-  
 210 posals are computed from the nuisance-only residual,  
 211 and each start is fitted both with and without the  $H_0$   
 212 albedo patch, so a colour explanation fitted first cannot  
 213 hide a dent.

214 Decision. The fit gain is  $\Delta = L_\Omega(H_0) - L_\Omega(H_1) \geq 0$ ,  
 215 as  $H_1$  nests  $H_0$ . We report  $D$  if  $\Delta > \tau$  and  $D = 0$  other-  
 216 wise, with  $\tau$  the split-conformal  $(1-\alpha)$  quantile of  $\Delta$  on  
 217 calibration negatives (healthy and colour-only blades),  
 218 which bounds the false-change rate on exchangeable  
 219 negatives by  $\alpha$ . We use a Charbonnier loss ( $\epsilon = 2/255$ )

| Split           | Blades | Healthy                                | Stain | Dent | Dent + stain |
|-----------------|--------|----------------------------------------|-------|------|--------------|
| Shader training | 13     | only for the renderer; never evaluated |       |      |              |
| Development     | 12     | 12                                     | 24    | 36   | 24           |
| Calibration     | 12     | 24                                     | 36    | 36   | 24           |
| Test (sealed)   | 30     | 30                                     | 60    | 90   | 60           |

Table 1. BladeDelta case counts. Each case has 7 rendered views; methods use 2 by default.

summed over  $\Omega$ ; a Student- $t$  noise model and a wider Charbonnier floor did not improve separation on development data.

#### 4. The BladeDelta Benchmark

Geometry. Healthy blades are public Rotor37 compressor geometries [4] (CC BY-SA 4.0), normalised to unit span and closed with a labelled artificial hub collar. Each mesh has 29,989 vertices. Shapes 0–12 were used during earlier development and only to train the shader. Shapes 13–24 form the development split, 25–36 the calibration split and 37–66 the sealed test split; 67–109 are held in reserve.

Damage. Each blade receives one of: no change, a colour-only patch (a soft darkened or rust-tinted albedo change), a dent, or a dent plus a colour patch (half overlapping). Dents follow one of three families — a cubic profile with a raised rim, a Gaussian, and an anisotropic elliptic profile (aspect 0.4–1) — with depth log-uniform in  $[0.0005, 0.006]$  blade spans (about 0.04–0.45 mm on a 74 mm Rotor37 blade) and radius in  $[0.04, 0.12]$ . Centres face at least one of seven cameras and lie at least  $\max(0.1, 1.2r)$  from the airfoil boundary.

Capture. Images are rendered with Blender Cycles (32 samples, denoised, AgX tone mapping,  $512 \times 512$ ) from seven cameras on a ring around the blade, under one of four conditions: nominal; camera jitter (yaw, pitch and roll each uniform in  $\pm 1.5^\circ$  per view); light jitter (each of three area lights  $\times [0.85, 1.15]$ ); or both. Methods receive only the nominal cameras and lighting. The default input is two views (60° and 120°); four- and seven-view settings are ablations.

Protocol. The split, case lists, metrics and hypotheses were hash-frozen before any test image was rendered; protocol changes are logged with dates. For each damaged case we record observability, the depth-weighted fraction of the dent visible in the input views.

CAD6 transfer split. To test transfer beyond Rotor37 we add six industrial blade designs supplied as STEP solids (airfoil spans 24–70 mm), with platforms, fillets and dovetail roots. The airfoil is the largest B-rep face (two faces on one blade); everything else is excluded from evidence like the Rotor37 collar. Each blade is normalised to unit airfoil span in the Rotor37 frame and receives 76 cases (456 in total): 30% dents centred where both input cameras see the surface, 15% dents hidden from both, 20% healthy, 15% stain only, 10% dent plus stain and 10% healthy with camera error alone ( $\pm 0.5^\circ$ ,  $\pm 1^\circ$ ,  $\pm 1.5^\circ$ ). Dents are sized in millimetres as an inspector would specify them: depth log-uniform in  $[0.1, 1.0]$  mm and radius in  $[1, 5]$  mm. The case list was hashed before rendering, and no component — shader, threshold or hyperparameter — was refit on this split.

Metrics. Relative local error is the RMS vertex error over the true dent support divided by that of the unchanged CAD (1: no better than CAD; 0: exact). We also report relative whole-airfoil error, depth error, centre error, detection, false physical change on negatives (any vertex moved more than  $2 \times 10^{-5}$ ) and abstention on unobservable dents, with 95% bootstrap intervals resampled by blade shape.

## 5. Experiments

Baselines. Unchanged CAD reports the healthy mesh. Discrete search re-implements our earlier method: image-guided proposals, a fixed cubic profile of radius 0.08 and depths  $\{0.0005, 0.001, 0.0015\}$ , at nominal pose. Free-form field fits a per-vertex normal displacement with Laplacian smoothness and  $\ell_1$  sparsity after the same pose and lighting fit. MLP deformation field replaces the per-vertex values with a coordinate MLP (positional encoding, two hidden layers), the continuous deformation field used by CAD-prior reconstruction. These two baselines reimplement the core idea of CADSplat [17] — a smooth deformation field fit to sparse posed views under a CAD prior — within our renderer; they are not the official model, whose code had not been released at the time of writing. 2D detector, lifted is a U-Net trained on (photo, healthy render) pairs of separate exposed blades to segment dents; detections are back-projected onto the mesh through the nominal cameras. All methods share the renderer, cameras and observations, and each is thresholded by the same calibration procedure on its own score.

Main results.

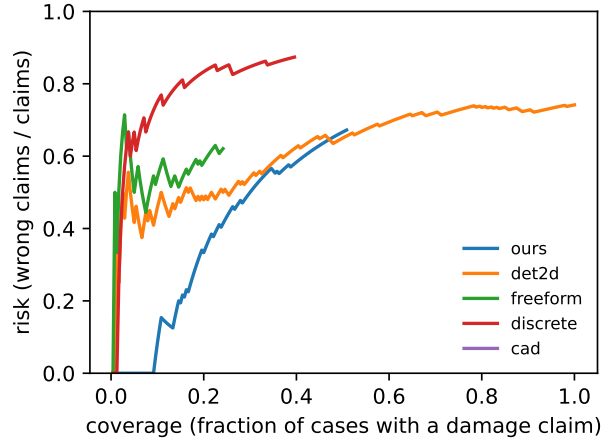


Figure 1. Risk (share of damage claims that are wrong) against coverage (share of cases with a claim) as the threshold varies, sealed test.

Risk-coverage.

Ablations.

Why recall is limited. Recall is set by dent depth and by the false-alarm budget, not by a failure to locate (Table 5). Every reported dent is in the right place, and 12/13 visible dents deeper than 0.2 mm are reported; most misses are shallower than 0.1 mm, where the change is about one grey level. In 6 of the 17 misses the dent hypothesis converged on the true location, but its evidence stayed below  $\tau = 1.52$ . The threshold is held up by appearance changes: stains account for 8 of the 9 false alarms, with stain-only fit gains up to 4.7 (5.6 for one healthy blade under camera error). Lowering  $\tau$  to 0.54 would reach 70% recall at a 20% false-alarm rate. The misses are therefore the price of the false-alarm guarantee: an inspector receives few spurious damage claims, and dents below about 0.1 mm are reported as unsupported rather than guessed.

Transfer to unseen blade designs. Table 6 applies every method, unchanged, to the six CAD6 blades (456 cases). Separation falls from 0.82 on the Rotor37 sealed test to 0.74 (transfer gap 0.08), with 94% of visible dents found, 67% false alarms and 33% abstention on hidden dents. The threshold was calibrated on Rotor37 only, so the false-alarm rate here measures how far the guarantee survives a change of geometry, not a property the method was fit to. Per blade (Table 7), dents are still ranked above negatives (best AUROC on blade 1 (24 mm, AUROC 0.98), worst on blade 3

| Method                             | AUROC $\uparrow$  | Found % $\uparrow$ | Located % $\uparrow$ | False alarm % $\downarrow$ | Abstain (hidden) % $\uparrow$ | Depth err. (mm) |
|------------------------------------|-------------------|--------------------|----------------------|----------------------------|-------------------------------|-----------------|
| Ours (v1, frozen)                  | 0.82 [0.71, 0.90] | 54 [39, 68]        | 54 [39, 68]          | 10 [4, 17]                 | 85 [77, 91]                   | 0.03            |
| without the evidence test          | 0.82 [0.71, 0.90] | 84 [68, 95]        | 70 [56, 84]          | 32 [21, 43]                | 45 [36, 54]                   | 0.03            |
| 2D detector, lifted                | 0.71 [0.63, 0.80] | 24 [12, 40]        | 16 [5, 30]           | 4 [1, 9]                   | 96 [93, 99]                   | —               |
| MLP deformation field <sup>†</sup> | 0.74 [0.65, 0.81] | 16 [6, 29]         | 11 [3, 22]           | 6 [1, 10]                  | 100 [100, 100]                | —               |
| Free-form field <sup>†</sup>       | 0.63 [0.54, 0.70] | 14 [4, 24]         | 11 [3, 20]           | 4 [1, 9]                   | 99 [97, 100]                  | —               |
| Discrete search                    | 0.53 [0.43, 0.65] | 5 [0, 13]          | 5 [0, 13]            | 6 [1, 10]                  | 98 [95, 100]                  | —               |
| Unchanged CAD                      | 0.50 [0.50, 0.50] | 0 [0, 0]           | 0 [0, 0]             | 0 [0, 0]                   | 100 [100, 100]                | —               |

Table 2. Sealed test (240 cases, 30 unseen Rotor37 blades). Found / located: visible dents ( $\geq 50\%$  visible in the two input views) reported / reported within 0.1 span of the true centre. Each method uses its own threshold, calibrated on the calibration split (split-conformal,  $\alpha = 0.05$ ); "without the evidence test" reports every fit. Errors are medians over reported dents. Mean [95% interval, resampling blades]. <sup>†</sup>Reimplementations of the core idea of CADSplat [17], not the official model.

| Views   | Observable dents | AUROC $\uparrow$ | Located % $\uparrow$ | Median local err. $\downarrow$ | Method                    | AUROC             | Found (%)   | Located (%) |
|---------|------------------|------------------|----------------------|--------------------------------|---------------------------|-------------------|-------------|-------------|
| 2 views | 10/60            | 0.847            | 60                   | 0.03                           | Ours, Rotor37 sealed test | 0.82 [0.71, 0.90] | 54 [39, 68] | 54 [39, 68] |
| 4 views | 17/60            | 0.777            | 41                   | 0.00                           | Ours                      | 0.74 [0.67, 0.84] | 94 [89, 98] | 70 [56, 84] |
| 7 views | 29/60            | 0.857            | 62                   | 0.47                           | 2D detector, lifted       | 0.86 [0.83, 0.89] | 42 [35, 49] | 36 [3, 40]  |
|         |                  |                  |                      |                                | MLP deformation field     | 0.73 [0.68, 0.81] | 84 [70, 97] | 54 [4, 68]  |
|         |                  |                  |                      |                                | Free-form field           | 0.60 [0.56, 0.66] | 61 [36, 84] | 37 [2, 50]  |
|         |                  |                  |                      |                                | Discrete search           | 0.65 [0.58, 0.72] | 16 [12, 18] | 13 [1, 26]  |
|         |                  |                  |                      |                                | Unchanged CAD             | 0.50 [0.50, 0.50] | 0 [0, 0]    | 0 [0, 0]    |

Table 3. More views make more dents observable and the evidence clearer.

| Variant                                  | AUROC $\uparrow$ | Located % $\uparrow$ | Method                                | AUROC             | Found (%)   | Located (%) |
|------------------------------------------|------------------|----------------------|---------------------------------------|-------------------|-------------|-------------|
| Full method                              | 0.855            | 79                   | Ours v1 (frozen), held-out blades 3–6 | 0.72 [0.65, 0.90] | 90 [85, 95] | 67 [5, 81]  |
| No camera (pose) correction              | 0.679            | 72                   | Ours v1.1, held-out blades 3–6        | 0.95 [0.93, 0.98] | 86 [77, 95] | 83 [7, 91]  |
| No stain (colour) model                  | 0.753            | 83                   | Ours, CAD6 leave-one-blade-out $\tau$ | 0.70 [0.63, 0.84] | 13 [3, 26]  | 6 [1, 13]   |
| Whole image instead of interior evidence | 0.685            | 64                   |                                       |                   |             |             |
| No score-test proposals                  | 0.420            | 46                   |                                       |                   |             |             |
| No low-frequency brightness field        | 0.846            | 72                   |                                       |                   |             |             |

Table 4. Ablations on the development subset (18 observable dents, 18 healthy or stained blades).

| True depth (mm)   | 2 views         | 2 views (same cases) | Blade 4 views | Airfoil span (mm) 7 views | Cases | Visible dents | AUROC | Found |
|-------------------|-----------------|----------------------|---------------|---------------------------|-------|---------------|-------|-------|
| < 0.1             | 1/10 (1 loc.)   | 0/4 (0 loc.)         | 1/5 (0 loc.)  | 3/36 (3 loc.)             | 76    | 31            | 0.98  | 100   |
| 0.1–0.2           | 7/14 (7 loc.)   | 1/2 (1 loc.)         | 2/6 (2 loc.)  | 7/36 (7 loc.)             | 76    | 31            | 0.72  | 100   |
| > 0.2             | 12/13 (12 loc.) | 4/4 (4 loc.)         | 4/6 (4 loc.)  | 8/31 (6 loc.)             | 76    | 31            | 0.64  | 97    |
| All visible dents | 20/37 (20 loc.) | 5/10 (5 loc.)        | 7/17 (6 loc.) | 18/39 (16 loc.)           | 76    | 31            | 0.66  | 94    |
|                   |                 |                      | 6             | 70                        | 76    | 31            | 0.93  | 84    |
|                   |                 |                      |               |                           |       |               | 0.86  | 87    |

Table 5. Visible test dents reported (located within 0.1 span) by true depth, at the frozen threshold. Below 0.1 mm a dent changes the image by about one grey level; the method declines rather than guess.

Table 6. Transfer to six unseen real blade designs (CAD6, 456 cases; dents 0.1–1.0 mm deep). Every method is frozen: shader trained on Rotor37 only, thresholds from the Rotor37 calibration split. Top row: ours on the Rotor37 sealed test, for the transfer gap. Brackets: 95% intervals resampling blades.

Table 7. Ours on each CAD6 blade, smallest to largest airfoil (frozen threshold).

334 (31 mm, AUROC 0.64)), but the frozen threshold fails  
 335 on the smaller blades: false alarms fall with airfoil span  
 336 (Spearman  $\rho = -0.88$ ,  $p = 0.02$ ). Across blades, air-  
 337 foil span and detection rate have Spearman  $\rho = -0.93$   
 338 ( $p = 0.01$ ; six blades, so weak evidence): 99% of visi-  
 339 ble dents were found on the three smaller blades and

88% on the three larger ones. Because dents are spec-  
 340 ified in millimetres, the same dent is relatively deeper  
 341 on a small blade; detection tracks depth relative to the  
 342 airfoil span ( $\rho = 0.10$ ) more strongly than depth in mil-  
 343 limetres ( $\rho = 0.02$ ), rising from 92% in the shallowest  
 344 relative-depth tertile to 97% in the deepest. Diagnosis  
 345 and fix (v1.1). On a healthy CAD6 capture whose im-  
 346

| Variant (CAD6, 48 cases) | AUROC             | Found (%)    | False Method (%) | AUROC             | Found (%)   | Location     |
|--------------------------|-------------------|--------------|------------------|-------------------|-------------|--------------|
| Full method              | 0.71 [0.62, 0.83] | 92 [83, 100] | 78 [56, 100]     | 0.79 [0.68, 0.88] | 54 [40, 69] | 46 [32, 100] |
| No camera correction     | 0.90 [0.81, 0.98] | 92 [83, 100] | 56 [33, 100]     | 0.80 [0.70, 0.88] | 52 [38, 67] | 46 [32, 100] |
| No stain model           | 0.73 [0.64, 0.84] | 96 [88, 100] | 89 [78, 100]     |                   |             |              |
| Random starts            | 0.45 [0.37, 0.56] | 38 [29, 46]  | 41 [28, 97]      |                   |             |              |
| Fixed grid of starts     | 0.50 [0.41, 0.62] | 33 [17, 54]  | 47 [19, 82]      |                   |             |              |
| Whole image              | 0.56 [0.44, 0.71] | 75 [58, 92]  | 61 [40, 71]      |                   |             |              |

Table 8. Ablations on CAD6 (frozen settings, fixed 48-case subset).

|                                 |             |
|---------------------------------|-------------|
| Objects / defects / negatives   | 6 / 22 / 31 |
| AUROC (defect vs. undamaged)    | 0.540       |
| Detected (leave-one-object-out) | 0/22        |
| Located within 15 mm            | 0/22        |
| False alarms                    | 1/31        |

Table 10. Real objects (SiM3D) scanned before and after damage; truth from the measured scans.

Physical validation (pending real validation).

Pending real validation. A  $2\times$  print of CAD6 blade 5 will be dented with steel balls (target depths 0.3, 0.7 and 1.2 mm), photographed with marker-based poses before and after denting, and measured by structured-light scan (or a 0.01 mm dial gauge). This section is reserved for those results; the rest of the paper does not depend on it.

Qualitative results and failures. Figure 2 shows typical outcomes, including a miss and a false alarm.

Real objects. No public dataset pairs real blade photographs with measured damage geometry, so we use SiM3D [7]: six wooden stools scanned with an industrial structured-light sensor before and after damage, with calibrated photographs and expert defect annotations. The healthy scan replaces the CAD model, the healthy photograph from the same scanner position serves as the reference after 2D alignment, and the measured scan difference is the truth; annotated regions with no measured shape change (paint, contamination) count as negatives. On these textured objects our method does not separate physical defects from undamaged areas (Table 10). The residual between sessions is dominated by wood-grain misregistration and specular changes that the shader, trained on 12 photographs per object, does not model. We report this as a failure case: the evidence test is only as reliable as the appearance model behind it.

## 6. Limitations and Conclusion

Our positive results use rendered images of public compressor geometry with synthetic damage. On real

age is identical to the reference, the silhouette pose fit moves a correct camera by about  $0.3^\circ$  and a spurious dent absorbs the misregistration; under real camera error, the image-space reference bias — large on unseen geometry because the shader was trained on Rotor37 only — stays fixed while the blade moves. v1.1 corrects both: it applies the reference correction to the silhouette as well as to shading, and it reprojects the shading correction through the surface so that it moves with the blade (identical to v1 at the nominal pose). Both changes were developed on CAD6 blades 1–2 only and frozen, with a threshold recalibrated on the Rotor37 calibration split, before blades 3–6 were run. On those held-out blades v1.1 raises AUROC from 0.72 to 0.95, cuts false alarms from 51% to 4%, locates 83% of visible dents (v1: 67%) and abstains on 98% of hidden dents (v1: 50%). Recalibrating the threshold on CAD6 negatives, leave-one-blade-out, gives 13% found at 4% false alarms: the cost of transferring a threshold across blade designs.

A pre-registered negative result (v2). Targeting the remaining false alarms, we pre-registered three changes and a rule on development blades: a smooth surface appearance field shared by both hypotheses (A), a dent-size prior (D), or both (E), adopted only if recall at matched false-alarm rate improved by at least five points. None did: D matched v1 (67% recall, AUROC 0.87 vs. 0.85), while A and E raised negative fit gains so much that recall fell to 6%; the extra appearance freedom helped the dent hypothesis as much as the null. The change that does reduce false alarms is v1.1’s reference correction of silhouette and shading (above), which also lowers the worst development false-alarm gains on Rotor37 (2.46 to 0.09).

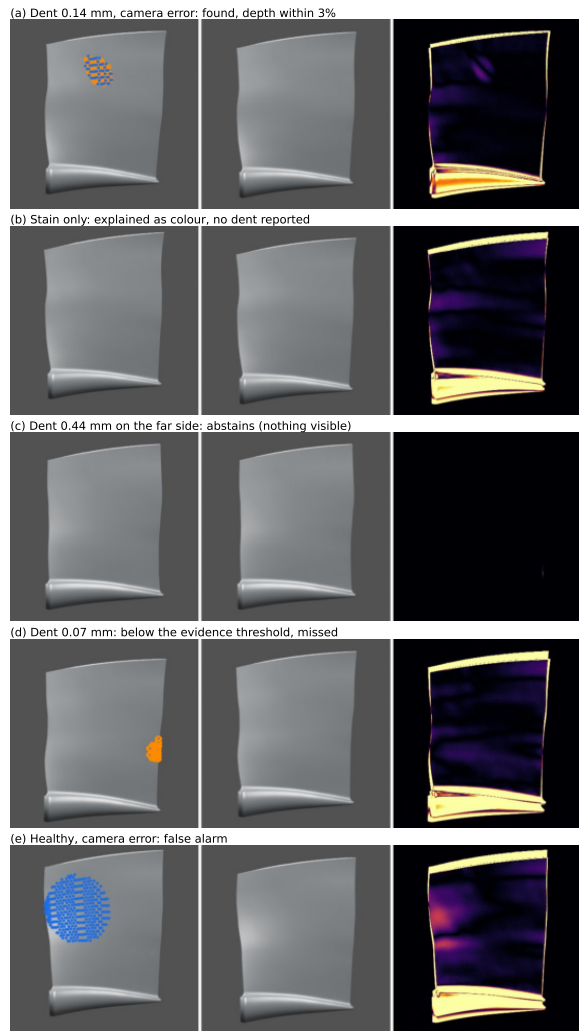


Figure 2. Sealed-test outcomes. (a) A 0.14mm dent under camera error is found and measured. (b) A stain is explained as an albedo change. (c) A dent facing away from both cameras yields no evidence and the method abstains. (d) A 0.07mm dent stays below the threshold. (e) A healthy blade under camera error is reported as dented, the residual failure mode.

scanned objects (SiM3D) the method did not separate physical defects from undamaged areas; real-blade validation with measured damage remains future work, and an appearance model that holds on real materials is its main prerequisite. The dent model has one dent per blade, and edge damage such as nicks and material loss is out of scope. The conformal guarantee holds for negatives exchangeable with the calibration set; a new camera rig or lighting setup requires recalibration. The method answers whether these views support a physical change; it does not certify airworthiness or replace

an inspector.

We framed blade damage estimation as a decision with a stated level of evidence: report a dent, with its location and depth, only when a dent explains the images significantly better than every non-physical explanation, and abstain otherwise. On rendered Rotor37 blades this cuts the false-claim rate of an unconstrained fit by two thirds (32% to 10%) while keeping most detectable dents, and it outperforms CAD-prior deformation fields and a lifted 2D detector, whose lack of an appearance model forces their thresholds so high that few dents survive. Recall is bounded by what the cameras see: almost all misses are shallower than 0.1–0.2mm, and seven views instead of two locate nearly three times as many dents at a lower false-alarm rate. Transfer to new blade designs exposed two assumptions that only held for the training family; making the healthy reference follow the silhouette and the camera removed the failure without retraining and without cost on Rotor37. Physical validation on a printed, dented and scanned blade (Section 5) is the next step.

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